

CCTV Camera Installation & CCTV Domain – Practical Tips Guide

This guide contains practical tips, installation methods, troubleshooting ideas, networking basics, and career guidance for beginners and intermediate professionals working in the CCTV and surveillance domain.

1. Basic CCTV Components

- Camera (Bullet, Dome, PTZ, IP Camera)
 - DVR/NVR for recording and monitoring
 - SMPS/Power Supply for stable power
 - Hard Disk for video storage
 - CAT6/Coaxial cable for transmission
 - RJ45 connectors and network switches
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2. Camera Placement Tips

- Install cameras at entry and exit points.
 - Avoid direct sunlight facing the lens.
 - Use wide-angle cameras for large halls.
 - Keep cameras 9–12 feet above ground for better visibility.
 - Avoid blind spots and overlapping issues.
 - Use waterproof cameras for outdoor areas.
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3. Networking Tips for IP Cameras

- Assign static IP addresses to cameras.
 - Use VLANs for surveillance traffic in large networks.
 - Check PoE switch power capacity before installation.
 - Test camera connectivity using ping command.
 - Use CAT6 cables for better speed and reliability.
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4. Cable Installation Best Practices

- Do not bend cables sharply.
- Keep power and data cables separated.
- Use conduit pipes for protection.

- Label both ends of every cable.
 - Use quality crimping tools for RJ45 connectors.
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5. DVR/NVR Configuration Tips

- Change default passwords immediately.
 - Enable motion detection recording.
 - Set proper date and time synchronization.
 - Configure backup recording schedules.
 - Enable remote access carefully with strong passwords.
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6. Troubleshooting Common Problems

- No video → Check power supply and cable connections.
 - Blurred image → Clean lens and adjust focus.
 - Camera offline → Verify IP address and switch port.
 - Hard disk error → Reformat or replace HDD.
 - Night vision issue → Check IR LEDs.
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7. Safety During Installation

- Use safety belts when working at heights.
 - Turn off power before wiring.
 - Use insulated tools.
 - Avoid water exposure to electrical components.
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8. Career Tips in CCTV Domain

- Learn basic networking and CCNA concepts.
 - Practice DVR/NVR configuration daily.
 - Understand firewall and switch basics.
 - Learn brands like Hikvision, Dahua, CP Plus, Aruba, Ruijie, and Grandstream.
 - Develop troubleshooting and communication skills.
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Quick Device Reference

Device	Purpose
Switch	Connects cameras and network devices
Router	Provides internet connectivity
PoE Switch	Provides power + data to IP cameras
Firewall	Secures CCTV network traffic
Access Point	Provides wireless connectivity